Tanay Malhotra

Embedded Firmware Engineer

RESUME_1390

Location: Kanpur, Uttar Pradesh, India | Email: tanay.malhotra.resume_1390@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 15 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 15 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: JTAG, SPI, Oscilloscope, ARM Cortex-M, I2C, STM32, UART, FreeRTOS, Git, C++, C, Embedded Linux

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | WillowGrid Labs | Jan 2021 - Present

- Lowered field-reset incidents by 15% with watchdog recovery and diagnostics
- Validated 680 hardware test vectors across protocol and timing corner cases
- Reduced embedded task jitter by 19% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | Evergrid Technologies | Jan 2016 - Dec 2020

- Reduced embedded task jitter by 39% using interrupt priority and queue tuning
- Improved battery runtime by 25% with peripheral and sleep-state optimization

Associate Embedded Firmware Engineer | KoraLink Innovations | Jul 2011 - Dec 2015

- Lowered field-reset incidents by 15% with watchdog recovery and diagnostics
- Validated 339 hardware test vectors across protocol and timing corner cases

PROJECTS

Industrial Sensor Firmware | Oscilloscope, FreeRTOS, ARM Cortex-M, Git

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Low-Power Metering Device | C++, I2C, STM32, ARM Cortex-M

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Eastgate Institute of Engineering, 2011
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

ARM Cortex-M Workshop (2022)